

LangChain: Agents

Outline:

- Using built in LangChain tools: DuckDuckGo search and Wikipedia
- Defining your own tools

```
In [1]: import os

from dotenv import load_dotenv, find_dotenv
_ = load_dotenv(find_dotenv()) # read local .env file

import warnings
warnings.filterwarnings("ignore")
```

Built-in LangChain tools

```
In [3]: #!/pip install -U wikipedia
```

```
In [4]: from langchain.agents.agent_toolkits import create_python_agent
from langchain.agents import load_tools, initialize_agent
from langchain.agents import AgentType
from langchain.tools.python.tool import PythonREPLTool
from langchain.python import PythonREPL
from langchain.chat_models import ChatOpenAI
```

```
In [5]: llm = ChatOpenAI(temperature=0)
```

```
In [6]: tools = load_tools(["llm-math","wikipedia"], llm=llm)
```

```
In [7]: agent= initialize_agent(
    tools,
    llm,
    agent=AgentType.CHAT_ZERO_SHOT_REACT_DESCRIPTION,
    handle_parsing_errors=True,
    verbose = True)
```

```
In [ ]: agent("What is the 25% of 300?")
```

```
In [ ]: question = "Tom M. Mitchell is an American computer scientist \
and the Founders University Professor at Carnegie Mellon University (CMU)\
what book did he write?"
result = agent(question)
```

Python Agent

```
In [9]: agent = create_python_agent(  
    llm,  
    tool=PythonREPLTool(),  
    verbose=True  
)
```

```
In [10]: customer_list = [ ["Harrison", "Chase"],  
    ["Lang", "Chain"],  
    ["Dolly", "Too"],  
    ["Elle", "Elem"],  
    ["Geoff", "Fusion"],  
    ["Trance", "Former"],  
    ["Jen", "Ayai"]  
]
```

```
In [ ]: agent.run(f""""Sort these customers by \  
last name and then first name \  
and print the output: {customer_list}""")
```

View detailed outputs of the chains

```
In [ ]: import langchain  
langchain.debug=True  
agent.run(f""""Sort these customers by \  
last name and then first name \  
and print the output: {customer_list}""")  
langchain.debug=False
```

Define your own tool

```
In [14]: #!/pip install DateTime
```

```
In [15]: from langchain.agents import tool  
from datetime import date
```

```
In [16]: @tool  
def time(text: str) -> str:  
    """Returns today's date, use this for any \  
    questions related to knowing today's date. \  
    The input should always be an empty string, \  
    and this function will always return today's \  
    date - any date mathematics should occur \  
    outside this function."""  
    return str(date.today())
```

```
In [17]: agent= initialize_agent(  
    tools + [time],  
    llm,  
    agent=AgentType.CHAT_ZERO_SHOT_REACT_DESCRIPTION,  
    handle_parsing_errors=True,  
    verbose = True)
```

Note:

The agent will sometimes come to the wrong conclusion (agents are a work in progress!).

If it does, please try running it again.

```
In [ ]: try:  
    result = agent("whats the date today?")  
except:  
    print("exception on external access")
```